

Class 1.6 — Control Box II

Meeting date: Sunday, July 19, 2026 · 3 hours · Mr. Holcombe's house

T-MINUS STRIP

T-2 WEEKS

Confirm 14 AWG silicone-jacketed wire (red + black, ~6 ft each), ring terminals, blade-fuse assortment (3/5/7.5/10/15/20 A) all in hand. Confirm the pre-

ORDER AHEAD · LEAD-TIME ITEMS
wired and on the central bench. Confirm the watt meter and fresh batteries.

- 14 AWG silicone-jacketed wire (red + black, ~6 ft each) — ~\$30 · by T-2 weeks
- Blade-fuse assortment (3 / 5 / 7.5 / 10 / 15 / 20 A) + ring terminals — ~\$25 · by T-2 weeks
- **12V SLA battery + charger + approved enclosure (gating from 1.5) — \$80 (covered by 1.5 order) · by Already in hand**
- Wire-strippers (gauge-correct) + ring-terminal crimper if not on hand — ~\$35 (one-time) · by T-1 week

STAGE IN THE ROOM

- Solder workstations: one iron per student (same as 1.5).
- Power-wire station: wire, terminals, crimper, stripper, heatshrink, lighter or heat gun.
- Control box from 1.5 in its tray; SeaMATE Guide queued at the next block.
- Multimeter at every two stations; watt meter on the central bench.
- Battery in approved enclosure, DISCONNECTED, on the central bench under the watt meter.
- Engineering notebook open at the fuse-selection page.

SET UP

- Bench layout: solo soldering stations + one dedicated power-wire station in clear sight of the advisor.
- Ventilation: fume extractor on, fan placed, before any iron heats.
- Safety: glasses at every station, first-aid kit visible, fire extinguisher within reach.
- Battery handling: enclosure on the central bench, two people verify DISCONNECTED before any other tool touches the bench.

CUE UP

- SeaMATE Guide on the iPad — bookmark to the next block of components past 1.5's stopping point.
- Reference: MATE "Creating your ROV's power wire" + "Connecting the Tether to the Control Box" presentations open in browser tabs.
- ELEC-NRD-003 (MATE safety rule on fuse sizing) printed for the engineering-notebook page.
- Slide deck 1.6 cued on the TV.

T-1 WEEK

Confirm 12V SLA battery, charger, and battery enclosure are in hand and the battery is fully charged. Verify the watt meter is calibrated and the

T-1 DAY

Re-stage the workstations: solo solder benches AND a dedicated power-wire station with the wire stripper, heatshrink, lighter, ring-terminal crimper. Lay out the multimeter and iPad at the right side.

- Closed-toe shoes.
- Notebook if keeping one (this is the night fuse-selection math lands in it).
- A water bottle.
- Hair tied back if long.

CONFIRM · GATING

- BATTERY DISCONNECTED at the start of class — two-person verify.
- Smoke test does NOT happen today. Pre-pool review at 1.7 is when we power up.
- Engineering notebook entry for fuse selection is non-negotiable before the board signs off.
- Pool prerequisites for 1.7 — lifeguard coverage, host rules in writing, photo releases — confirmed THIS WEEK, not at the pool deck.

PREP OWNERS

- Advisor — Order wire, terminals, fuses, crimper · by T-2 weeks
- Advisor — Stage power-wire station, test multimeter · by T-1 day
- CEO (or designee) — Confirm attendance + battery is charged · by T-1 day
- CFO (or designee) — Bring engineering notebook + ELEC-NRD-003 printout · by T-0
- Advisor + 1 parent — Confirm 1.7 pool prerequisites — lifeguard, host rules, photo releases · by T-0 of 1.6 (gating 1.7)

T-0 (MEETING DAY)

Doors at 2:45 PM, brief at 3:00 sharp. Battery in its enclosure on the bench, DISCONNECTED. Safety glasses at every station. Fume extractor running before any iron heats.